

Machine learning researcher and engineer

SKILLS

- **Tech:** Python (TensorFlow, PyTorch, Transformers, Optuna, scikit-learn, pandas), C, Matlab, Linux, Git, Docker, AWS
- **Communication:** French (native), English (fluent), Japanese (JLPT N2)

PROFESSIONAL EXPERIENCE

- Algorithm researcher, Brainomix** (Oxford, UK) Dec 2022–current
- Developed and deployed a respiratory phase classifier for chest CT using shape features from deep learning-based trachea/lung segmentations, supporting in-house imaging biomarker reliability. Investigated model explainability by examining feature importance, PDPs and ICE plots.
 - Conducted survival analysis using random survival forests and Cox proportional hazards models to assess segmentation quality and biomarker predictive power. Improved ML validation workflows in CI/CD by automating survival analysis and reducing mean runtime by 11%.
 - Improved trachea/lung segmentation postprocessing using a level-set-based method, reducing worst-case lung volume estimation error by 26%.
 - Explored denoising algorithms such as guided filtering for robust airway segmentation, improving Dice score by 0.055 at specific noise levels
- Visiting researcher, Sony Computer Science Laboratories** (Tokyo / remote) June–Nov 2022
- Explored NLP methods, e.g., transformer-based models (BERT, MPNet) and LSTMs with GloVe embeddings (in [TensorFlow/PyTorch](#)), for fake-news detection, ESG risk-score regression, and analysis of fund-report alignment with ESG criteria to support Nomura Asset Management
 - Designed an interpretable method for fund-level greenwashing detection by deriving projection-distance formulas to compare neural embeddings of internal sustainability reports and external news in a subspace defined using ESG criteria
 - Developed an optical-flow-inspired method to estimate time-varying loadings in security factor models and trained a VAE to learn and visualize low-dimensional fund-exposure embeddings from per-stock active weights and loadings, supporting GPIF, a major Japanese pension fund
- Computer vision research intern, Idemia, Research and Technology Unit** (Île-de-France, France) Apr–Sept 2016
- Prototyped an end-to-end OCR pipeline in [C](#) for degraded text images—segmentation, character recognition, and shortest-path graph decoding using Viterbi’s algorithm to select the most likely string—achieving 13.4% higher accuracy than Tesseract 3.04, an open-source OCR engine
 - Designed a method for character over-segmentation (i.e., candidate cut generation) based on Dijkstra’s algorithm, modeling the image as a graph with pixels as nodes and edge weights derived from intensities and heuristics; reduced time complexity via a binary-search-style strategy
 - Implemented a character-level neural-network classifier from scratch (~50k parameters) as part of the OCR pipeline and trained it on ~20k synthetic character images; improved accuracy by ~5% by incorporating a class-conditional Bayesian aspect-ratio prior
- Physics research intern, Schlumberger, Engineering department** (Kanagawa, Japan) Jan–Feb 2014 and June–Aug 2014
- Modeled fluid flow with PDEs and solved them via harmonic/asymptotic analysis to assess vibration-sensor feasibility for reservoir evaluation

EDUCATION

- M.Sc. and Ph.D. in Bioengineering, The University of Tokyo** (Japan) Sept 2016–Sept 2021
- Derived implementations of online training algorithms for RNNs with improved accuracy and complexity; applied them to 3D marker-position forecasting for latency compensation in radiotherapy, achieving competitive results with less training data than prior deep-learning approaches
 - Developed a data-efficient, modular, and interpretable image-forecasting algorithm for MR-guided radiotherapy, combining deformable image registration, PCA-based motion modeling, and temporal dynamics prediction with online-trained RNNs and transformers in PyTorch/Matlab
 - [Published 4 articles in Q1 CS journals](#) and [1 article chosen as an Editors’ Pick in "Towards Data Science"](#); the 2 most recent papers extend my Ph.D. work with new algorithms/data and experiments (e.g., ablation/statistical analyses) conducted during [independent research \(2022–2026\)](#)
 - Presented my research at 12 conferences in Japan (2 awards), 1 workshop at Seoul National University, and in [1 YouTube video \(~450k views\)](#)
 - Secured competitive scholarships & grants from the Japan Student Services Organization, the Epson International Scholarship Foundation, and UTokyo, e.g., the Doctoral Student Special Incentives Program Type A (top 20 2nd-year Ph.D. applicants at the Graduate School of Engineering)
 - Independently completed 9 additional M.Sc.-level remote courses in math, alongside UTokyo studies, at Aix-Marseille Université (2017–2019)
- M.Sc. in Engineering** (Diplôme d’ingénieur) and **M.Sc. in Signal & Image Processing, Centrale Lille/Centrale Méditerranée** (France) 2013–2016
- General engineering and management curriculum followed by a specialization track in AI and computer vision research; [valedictorian award](#)
 - Implemented a [classifier for graph-structured data](#) combining generalized likelihood-ratio testing with graph Fourier transforms; applied it to Alzheimer’s disease diagnosis from PET images by modeling brain regions as graph nodes (3-month research project at Fresnel Institute)
 - Implemented a [classifier for graph-structured data](#) combining generalized likelihood-ratio testing with graph Fourier transforms; applied it to Alzheimer’s disease diagnosis from PET images by modeling brain regions as graph nodes (F1 score: 0.85, 81 patients and 61 healthy subjects)
 - Built/programmed autonomous robots using C/Arduino (e.g., proportional control) for the French Robotics Cup (2-year project, 6-student team)
- B.Sc. in Mathematics, Physics and Computer Science** (Classes préparatoires), **Lycée Louis-le-Grand** (Paris, France) 2010–2013
- Highly selective undergraduate science program preparing for nationwide competitive examinations for entry to the best French universities